Linux OS and Scripting
Assignment 3

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Subject: Linux OS and Scripting

Linux Filesystem

What is file system?

A file system is the method and data structure that an operating system uses to control how data is stored and retrieved. Without a file system, information placed in a storage medium would be one large body of data with no way to tell where one piece of information stops and the next begins. The Linux File System is a structured method of storing and organizing data on a Linux machine. It arranges files in a hierarchical directory format starting from the root directory (/).

What do you mean by storage configuration in Linux?

Storage configuration in Linux is the process of setting up hard drives and partitions so the operating system can save and read files.

Exercise 1 – Identify Your Linux Filesystem

Explore the storage configuration of your Linux system.

Determine:

1. The storage devices available.

```
msis@master:/$ lsblk
NAME        MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
loop0         7:0    0     4K  1 loop /snap/bare/5
loop1         7:1    0    74M  1 loop /snap/core22/2411
loop2         7:2    0    74M  1 loop /snap/core22/2437
loop3         7:3    0  66.8M  1 loop /snap/core24/1587
loop4         7:4    0  11.1M  1 loop /snap/firmware-updater/167
loop5         7:5    0  66.8M  1 loop /snap/core24/1643
loop6         7:6    0 614.5M  1 loop /snap/gnome-46-2404/164
loop7         7:7    0 260.3M  1 loop /snap/firefox/8763
loop8         7:8    0 257.6M  1 loop /snap/firefox/8664
loop9         7:9    0 606.1M  1 loop /snap/gnome-46-2404/153
loop10        7:10   0 531.5M  1 loop /snap/gnome-42-2204/263
loop11        7:11   0  16.5M  1 loop /snap/firmware-updater/226
loop12        7:12   0  11.8M  1 loop /snap/snap-store/1390
loop13        7:13   0   395M  1 loop /snap/mesa-2404/1165
loop14        7:14   0 531.4M  1 loop /snap/gnome-42-2204/247
loop15        7:15   0   402M  1 loop /snap/mesa-2404/1839
loop16        7:16   0  15.7M  1 loop /snap/snap-store/1367
loop17        7:17   0  50.1M  1 loop /snap/snapd/27710
loop18        7:18   0  91.7M  1 loop /snap/gtk-common-themes/1535
loop19        7:19   0  50.1M  1 loop /snap/snapd/27591
loop20        7:20   0   828K  1 loop /snap/snapd-desktop-integration/391
loop21        7:21   0 219.9M  1 loop /snap/thunderbird/1201
loop22        7:22   0   828K  1 loop /snap/snapd-desktop-integration/387
loop23        7:23   0 221.1M  1 loop /snap/thunderbird/1228
nvme0n1     259:0    0 953.9G  0 disk
├─nvme0n1p1 259:1    0   200M  0 part /boot/efi
├─nvme0n1p2 259:2    0    16M  0 part
├─nvme0n1p3 259:3    0 299.2G  0 part
├─nvme0n1p4 259:4    0   811M  0 part
├─nvme0n1p5 259:5    0   400G  0 part
├─nvme0n1p6 259:6    0    16G  0 part
└─nvme0n1p7 259:7    0 237.7G  0 part /
```

This command shows disks, partitions, sizes and mount point in a tree structure

Disk: The physical storage drive hardware (like an SSD or HDD) connected to your computer.

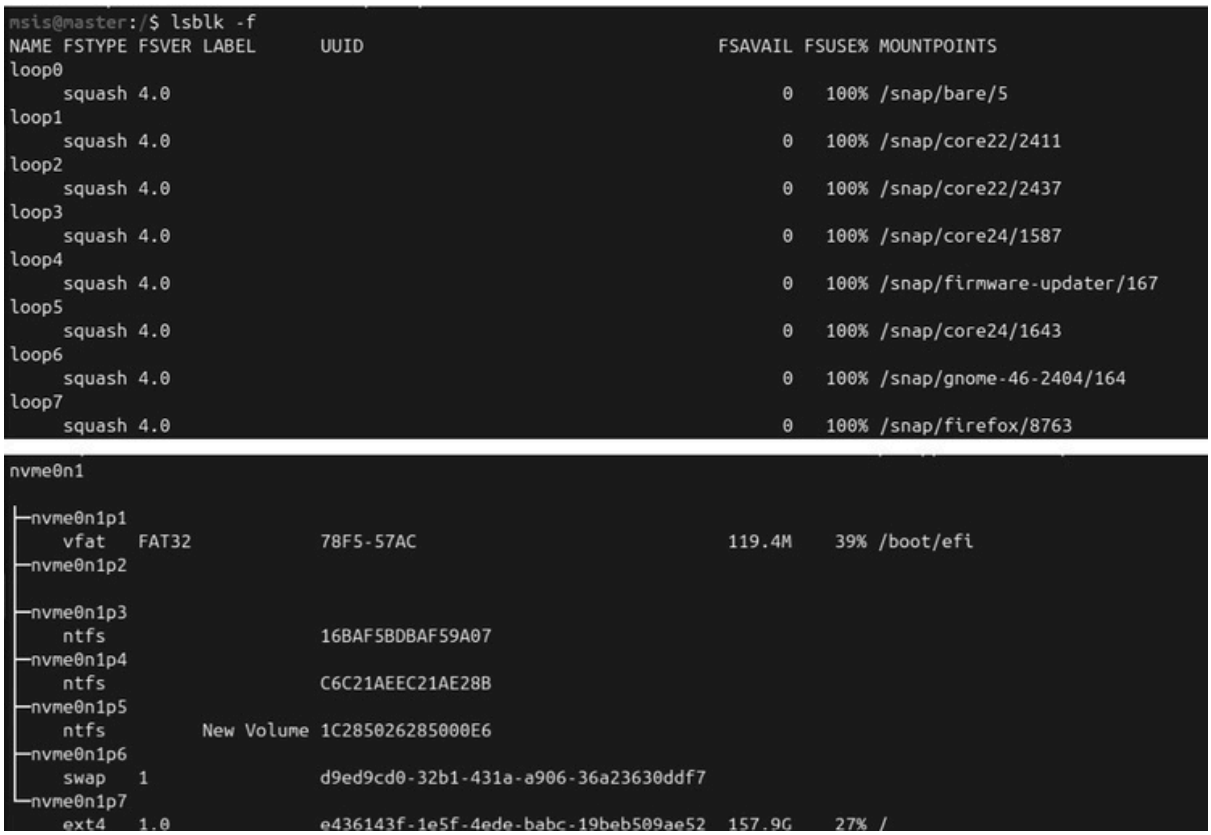
Partition: A logical section or isolated slice of a disk used to separate data.

Size: The total data capacity allocated to a physical disk or logical partition.

Mount Point: The directory folder where a partition is attached to be viewed and accessed.

Real-world use: An administrator uses lsblk to identify whether a server has disks such as /dev/sda, /dev/nvme0n1 , USB drives or additional storage

2. The partitions available on each device.



```
msis@master:/$ lsblk -f
NAME FSTYPE FSVER LABEL      UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
loop0
     squash 4.0                                                         0   100% /snap/bare/5
loop1
     squash 4.0                                                         0   100% /snap/core22/2411
loop2
     squash 4.0                                                         0   100% /snap/core22/2437
loop3
     squash 4.0                                                         0   100% /snap/core24/1587
loop4
     squash 4.0                                                         0   100% /snap/firmware-updater/167
loop5
     squash 4.0                                                         0   100% /snap/core24/1643
loop6
     squash 4.0                                                         0   100% /snap/gnome-46-2404/164
loop7
     squash 4.0                                                         0   100% /snap/firefox/8763
nvme0n1
├─nvme0n1p1
│    vfat   FAT32           78F5-57AC                            119.4M    39% /boot/efi
├─nvme0n1p2
│
├─nvme0n1p3
│    ntfs                   16BAF5BDBAF59A07
├─nvme0n1p4
│    ntfs                   C6C21AEEC21AE28B
├─nvme0n1p5
│    ntfs         New Volume 1C285026285000E6
├─nvme0n1p6
│    swap   1               d9ed9cd0-32b1-431a-a906-36a23630ddf7
└─nvme0n1p7
     ext4   1.0             e436143f-1e5f-4ede-babc-19beb509ae52  157.9G    27% /
```

Adds filesystem type, UUID, and mount point.

Real-world use: Useful when troubleshooting boot problems or determining which filesystem is mounted on a particular partition.

3. The filesystem type used by each partition.

```
msis@master:/$ df -T
Filesystem     Type     1K-blocks     Used Available Use% Mounted on
tmpfs          tmpfs      1603000     3076   1599924   1% /run
/dev/nvme0n1p7 ext4     244179276 66106384 165596428  29% /
tmpfs          tmpfs      8014996        4   8014992   1% /dev/shm
tmpfs          tmpfs         5120        8      5112   1% /run/lock
efivarfs       efivarfs       246      178        64  74% /sys/firmware/efi/efiv
ars
tmpfs          tmpfs      8014996        0   8014996   0% /run/qemu
/dev/nvme0n1p1 vfat        200704    78476    122228  40% /boot/efi
tmpfs          tmpfs      1602996     2556   1600440   1% /run/user/1000
```

Displays filesystem type, total size, used space, available space, and mount point.

Real-world use: A security analyst checks whether a filesystem is nearly full because malware logs, large dumps, or suspicious files may consume disk space.

4. The device/partition on which `/` is located.

```
msis@master:/$ df -Th /
Filesystem     Type  Size  Used Avail Use% Mounted on
/dev/nvme0n1p7 ext4  233G   64G  158G  29% /
```

This directly tells you the filesystem containing /.

Real-world use: Useful when determining which disk/partition contains the operating system.

5. The filesystem type used for `/`.

```
msis@master:/$ findmnt /
TARGET
  SOURCE         FSTYPE OPTIONS
/ /dev/nvme0n1p7 ext4   rw,relatime
```

6. The total, used, and available storage space.

Device/Partition	Filesystem type	Size	Mount Point
nvme0n1p1	vfat	200M	/boot/efi
nvme0n1p2	-	16M	-
nvme0n1p3	ntfs	299.3G	-
nvme0n1p4	ntfs	811M	-
nvme0n1p5	ntfs	400G	-
nvme0n1p6	swap	16G	-
nvme0n1p7	ext4	237.7G	/

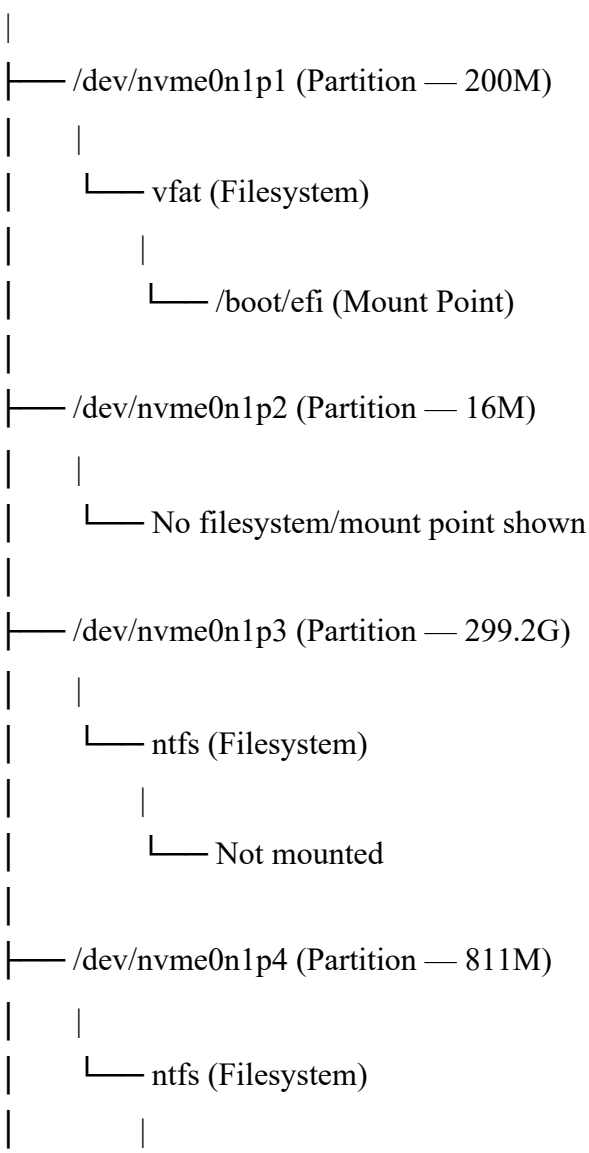
Exercise 2 – Disk → Partition → Filesystem → Mount Point

Explain in 2–3 sentences the relationship between these four components.

Relationship: A disk is the physical or virtual storage device. A partition is a defined section of that disk, a filesystem organizes data within the partition, and a mount point is the location in the Linux directory tree where that filesystem becomes accessible.

Real-world cybersecurity use: During forensic investigation, identifying this chain helps an investigator determine exactly where evidence is physically/virtually stored.

Using the information obtained in Exercise 1, draw the storage organization of your own Linux system.



```
│           └── Not mounted
│
├── /dev/nvme0n1p5 (Partition — 400G)
│     │
│     └── ntfs (Filesystem)
│           │
│           └── Not mounted
│
├── /dev/nvme0n1p6 (Partition — 16G)
│     │
│     └── swap
│           │
│           └── [SWAP]
│
└── /dev/nvme0n1p7 (Partition — 237.7G)
      │
      └── ext4 (Filesystem)
            │
            └── / (Root Mount Point)
```

Exercise 3 – Explore the Root Filesystem

Start from your current location.

1. Determine your current working directory.

```
msis@master:~$ pwd
/home/msis
```

pwd-print working directory

2. Navigate to the root directory.

```
msis@master:~/Day1$ cd /
msis@master:/$
```

3. Display the contents of the root directory.

```
msis@master:/$ ls
bin                dev   lib64              mnt       root              snap      tmp
bin.usr-is-merged  etc   lib.usr-is-merged  nfs_share run               srv       usr
boot               home  lost+found         opt       sbin              swap.img  var
cdrom              lib   media              proc      sbin.usr-is-merged sys
```

4. Identify all directories directly available under `/`.

```
msis@master:/$ ls -ld /
drwxr-xr-x 24 root root 4096 Aug 11 13:50 /
```

5. Determine whether hidden entries are present.

```
msis@master:/$ ls -la /
total 4194408
drwxr-xr-x  24 root root       4096 Aug 11 13:50 .
drwxr-xr-x  24 root root       4096 Aug 11 13:50 ..
lrwxrwxrwx   1 root root          7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x   2 root root       4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x   4 root root       4096 Aug 25 09:58 boot
dr-xr-xr-x   2 root root       4096 Aug  5  2025 cdrom
drwxr-xr-x  20 root root       4960 Sep  1 17:44 dev
drwxr-xr-x 155 root root      12288 Sep  1 18:03 etc
drwxr-xr-x   8 root root       4096 Sep  1 10:07 home
lrwxrwxrwx   1 root root          7 Apr 22  2024 lib -> usr/lib
lrwxrwxrwx   1 root root          9 Apr 22  2024 lib64 -> usr/lib64
drwxr-xr-x   2 root root       4096 Apr  8  2024 lib.usr-is-merged
drwx------   2 root root      16384 Apr 21 20:26 lost+found
drwxr-xr-x   4 root root       4096 Apr 24 16:19 media
drwxr-xr-x   3 root root       4096 Aug 25 16:33 mnt
drwxrwxrwx   2 root root       4096 Aug 11 13:50 nfs_share
drwxr-xr-x   4 root root       4096 Aug  7 10:52 opt
dr-xr-xr-x 536 root root          0 Sep  1 17:44 proc
drwx------   7 root root       4096 Sep  1 16:08 root
drwxr-xr-x  45 root root       1280 Sep  1 18:00 run
lrwxrwxrwx   1 root root          8 Apr 22  2024 sbin -> usr/sbin
drwxr-xr-x   2 root root       4096 Mar 31  2024 sbin.usr-is-merged
drwxr-xr-x  16 root root       4096 Apr 24 16:17 snap
drwxr-xr-x   2 root root       4096 Aug  5  2025 srv
-rw-------   1 root root 4294967296 Apr 21 20:29 swap.img
dr-xr-xr-x  13 root root          0 Sep  1 17:44 sys
drwxrwxrwt  20 root root       4096 Sep  1 19:45 tmp
drwxr-xr-x  12 root root       4096 Aug  5  2025 usr
drwxr-xr-x  14 root root       4096 Apr 21 20:31 var
```

Exercise 4 – Linux Filesystem Hierarchy Investigation

Investigate the following locations:

/home

/root

/etc

/var

/var/log

/tmp

/usr

/boot

/dev

/proc

For each location:

1. Examine its contents.

2. Identify at least three entries, where available.

3. Determine the likely purpose of the directory.

Directory	Example	Purpose
/home	claude/, ubuntu/	Personal data and settings for each non-root user account.
/root	.bashrc, .config/, .profile, .ssh/	Home directory of the root (superuser) account, kept separate from /home.
/etc	.java/, .pwd.lock, ImageMagick-6/, LatexMk, PackageKit/	System-wide configuration files for the OS, network, users, and services.
/var	backups/, cache/, lib/, local/, lock -> /run/lock	Variable data that grows or changes while the system runs.
/var/log	README, alternatives.log, apt/, bootstrap.log, btmp	Log files for the system and applications.
/tmp	hsperfdata_root/, node-compile-cache/, phantomjs/	Temporary files, world-writable, cleared on reboot.
/usr	bin/, etc/, games/, include/, lib/	Bulk of installed user-space programs, libraries, and docs.

/boot	(empty in this container)	Kernel image and bootloader config on a normal machine.
/dev	autofs, console, cpu/, cpu_dma_latency, fd -> /proc/self/fd	Device files representing hardware and virtual devices.
/proc	numbered PID dirs, cpuinfo, meminfo	Virtual filesystem exposing live kernel and process info.

Exercise 5 – Find Your Home Directory

Without assuming its location:

1. Determine your username.

```
msis@master:/$ whoami
msis
```

2. Determine your home directory.

```
msis@master:/$ echo "$HOME"
/home/msis
```

3. Navigate to your home directory.

```
msis@master:/$ cd home
msis@master:/home$
```

4. Display its absolute path.

```
msis@master:/home$ pwd
/home
```

5. List its normal contents.

```
msis@master:/home$ ls
administrator  alice  bob  charlie  clusteruser  msis
```

6. **List all contents, including hidden files.**

```
msis@master:/home$ ls -la
total 32
drwxr-xr-x  8 root          root          4096 Sep  1 10:07 .
drwxr-xr-x 24 root          root          4096 Aug 11 13:50 ..
drwxr-x---  6 administrator administrator 4096 Jun 19 15:28 administrator
drwxr-x---  2 alice         alice         4096 Sep  1 10:07 alice
drwxr-x---  2 bob           bob           4096 Sep  1 10:07 bob
drwxr-x---  2 charlie       charlie       4096 Sep  1 10:07 charlie
drwxr-x---  6 clusteruser   clusteruser   4096 Aug  7 12:42 clusteruser
drwxr-x--- 36 msis          msis          4096 Sep  1 09:37 msis
```

Exercise 6 – Hidden Files Investigation

Examine your home directory.

Compare:

ls

ls -a

Identify at least 2 hidden files/directories.

```
msis@master:/home$ ls -la
total 32
drwxr-xr-x  8 root          root          4096 Sep  1 10:07 .
drwxr-xr-x 24 root          root          4096 Aug 11 13:50 ..
drwxr-x---  6 administrator administrator 4096 Jun 19 15:28 administrator
drwxr-x---  2 alice         alice         4096 Sep  1 10:07 alice
drwxr-x---  2 bob           bob           4096 Sep  1 10:07 bob
drwxr-x---  2 charlie       charlie       4096 Sep  1 10:07 charlie
drwxr-x---  6 clusteruser   clusteruser   4096 Aug  7 12:42 clusteruser
drwxr-x--- 36 msis          msis          4096 Sep  1 09:37 msis
```

1. **How does Linux identify a hidden file?**

   A filename beginning with . is normally treated as hidden

2. **Why does a normal `ls` not display these files?**

   Ls hides dot-prefixed entries by default.

3. **What do `.` and `..` mean?**

   . —> current directory

   .. —> parent directory

4. **Can a hidden file still be accessed if its name is known?**

   Yes

5. **From a cybersecurity perspective, why should hidden files not automatically be considered safe or malicious?**

   Hidden does not mean safe.

   For example, an attacker might place malicious scripts or persistence mechanisms in hidden locations such as: ~/.config/ ~/.ssh/ ~/.local/

Exercise 7 – Build a Small Filesystem Hierarchy

Explore: mkdir
Hint: you can create this hierarchy in a single command
Create the following files:

cyber_lab/evidence/network/traffic.txt

cyber_lab/evidence/system/processes.txt

cyber_lab/logs/security.log

cyber_lab/scripts/check.sh

cyber_lab/reports/investigation.txt

```
mkdir -p cyber_lab/evidence/network\
      cyber_lab/evidence/system \
      cyber_lab/logs \
      cyber_lab/scripts \
      cyber_lab/reports
```

```
msis@master:~$ tree cyber_lab
locales-launch: Data of en_US locale not found, generating, please wait...
cyber_lab
├── evidence
│   ├── network
│   │   └── traffic.txt
│   └── system
│       └── processes.txt
├── logs
│   └── security.log
├── reports
│   └── investigation.txt
└── scripts
    └── check.sh

12 directories, 0 files
```

Exercise 8 – Absolute and Relative Paths

Assume your current directory is:

~/cyber_lab

Perform the following:

1. Navigate to `evidence/network` using a relative path.

2. Navigate to `/etc` using an absolute path.

3. Return to your home directory using the shortest possible command.

4. Navigate to `cyber_lab/reports`.

5. Move one directory upward.

6. Display the current directory after every operation.

```
msis@master:~$ cd cyber_lab
msis@master:~/cyber_lab$ pwd
/home/msis/cyber_lab
msis@master:~/cyber_lab$ #1
msis@master:~/cyber_lab$ cd evidence/network
msis@master:~/cyber_lab/evidence/network$ pwd
/home/msis/cyber_lab/evidence/network
msis@master:~/cyber_lab/evidence/network$ #2
msis@master:~/cyber_lab/evidence/network$ cd /etc
msis@master:/etc$ pwd
/etc
msis@master:/etc$ #3
msis@master:/etc$ cd ~
msis@master:~$ #4
msis@master:~$ cd ~/cyber_lab/reports
msis@master:~/cyber_lab/reports$ pwd
/home/msis/cyber_lab/reports
msis@master:~/cyber_lab/reports$ #5
msis@master:~/cyber_lab/reports$ cd ..
msis@master:~/cyber_lab$ pwd
/home/msis/cyber_lab
```

What is an absolute path?

Absolute path: It's the full address of a file or folder, starting all the way from /. No matter where you currently are in the terminal, it always points to the same place — like /etc/passwd. **What is a relative path?**

Relative path: It's a shortcut path based on where you're standing right now. It only makes sense from your current folder so reports means different things depending on where you type it from

What is the difference between `/` and `~`?

Difference between / and ~: / is the very top of the whole filesystem — everything on the system lives under it. ~ is just a quick way to say "my home folder" (like /root or /home/alice) it's personal to whoever's logged in, not a fixed location like /

Exercise 9 – Identify File Types

Investigate the following:

/etc/passwd

/etc

/bin/bash

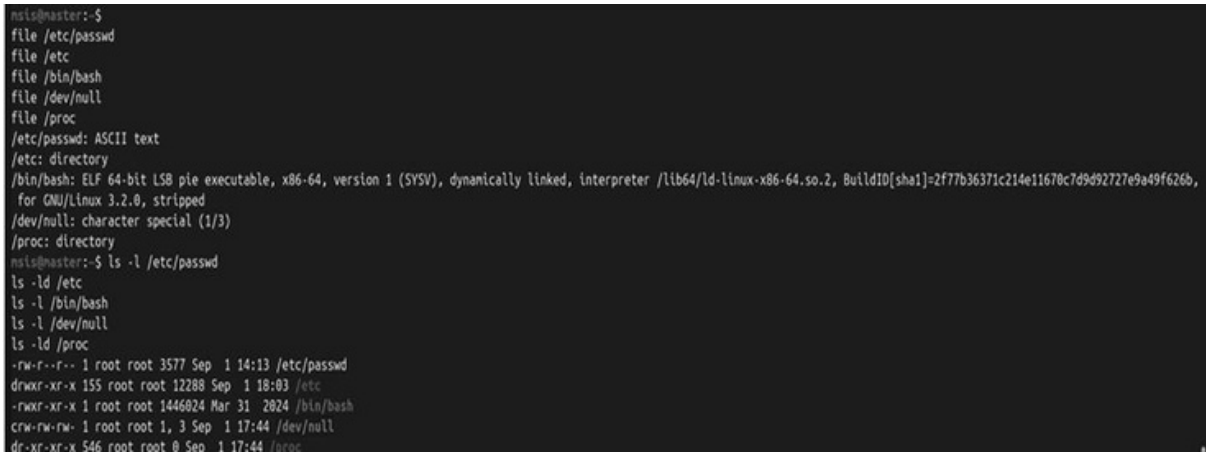
/dev/null

/proc

Also investigate the files created in Exercise 7.

file <filename>

ls -l <filename>



Exercise 10 – Examine File Metadata

Use:

~/cyber_lab/reports/investigation.txt

Add a small amount of text to the file.

Examine it using:

ls -l

stat investigation.txt

Identify:

* Filename

* File size

* File type
* Owner
* Group
* Inode number
* Access time
* Modification time
* Change time

```
msis@master:~/cyber_lab/reports$ ls -l investigation.txt
-rw-rw-r-- 1 msis msis 28 Sep  2 13:37 investigation.txt
msis@master:~/cyber_lab/reports$ stat investigation.txt
  File: investigation.txt
  Size: 28              Blocks: 8          IO Block: 4096   regular file
Device: 259,7   Inode: 12192134    Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/    msis)   Gid: ( 1000/    msis)
Access: 2026-09-02 13:37:55.915790335 +0530
Modify: 2026-09-02 13:37:49.817579321 +0530
Change: 2026-09-02 13:37:49.817579321 +0530
 Birth: 2026-09-02 13:37:49.816697274 +0530
```

1. What is metadata?

Metadata is information about a file, such as its size, owner, permissions, and timestamps.

2. Is metadata the same as file content?

No. Metadata describes the file, while content is the actual data inside the file.

3. Which timestamp changes when file contents are modified?

The Modification Time (mtime) changes when the file content is modified.

Exercise 11 – Explore `/etc`

Investigate:

/etc

Locate:

passwd
group
hostname
hosts

```
msis@master:~/cyber_lab/reports$ ls -l /etc/passwd
-rw-r--r-- 1 root root 3577 Sep  1 14:13 /etc/passwd
msis@master:~/cyber_lab/reports$ ls -l /etc/group
-rw-r--r-- 1 root root 1433 Sep  1 18:03 /etc/group
msis@master:~/cyber_lab/reports$ ls -l /etc/hostname
-rw-r--r-- 1 root root 7 Aug  7 11:24 /etc/hostname
msis@master:~/cyber_lab/reports$ ls -l /etc/hosts
-rw-r--r-- 1 root root 606 Aug  7 11:32 /etc/hosts
```

1. What kind of information is primarily stored under `/etc`?

/etc mainly stores system-wide configuration files.

2. What information does `/etc/passwd` contain?

It contains user account information such as usernames, user IDs, home directories, and login shells.

3. What information does `/etc/hostname` contain?

It contains the name (hostname) of the Linux system.

4. Why is `/etc` security-sensitive?

It is security-sensitive because it contains important system and security configuration files.

Exercise 12 – Explore Linux Log Storage

Investigate:

/var/log

Identify:

Available log files

Command: find /var/log -maxdepth 1 -type d

Available log directories

Command: find /var/log -maxdepth 1 -type d

Recently modified entries

Command: ls -lt /var/log | head

Entries you can read

Command: find /var/log -type f -readable

Entries you cannot read as a normal user

Command: find /var/log -type f ! -readable

1. Why does Linux maintain logs?

Linux maintains logs to record system, application, and security events.

2. Why are logs useful during incident investigation?

Logs help investigators determine what happened, when it happened, and who or what was involved.

3. Why might access to some logs be restricted?

Some logs contain sensitive security information, so access is restricted to prevent unauthorized access or modification.

Exercise 13 – Investigate `/tmp`

Explore `/tmp` without modifying files belonging to other users.

Determine:

1. Whether files/directories currently exist.

find /tmp -maxdepth 1 -type f

find /tmp -maxdepth 1 -type d

2. Whether hidden files exist.

la -la /tmp or

```
msis@master:~/cyber_lab/reports$ find /tmp -maxdepth 1 -name ".*"
/tmp/.X11-unix
/tmp/.X1025-lock
/tmp/.ICE-unix
/tmp/.X1024-lock
/tmp/.font-unix
/tmp/.XIM-unix
/tmp/.X0-lock
/tmp/.X1-lock
```

3. Who owns some of the entries.

ls-lah/tmp

```
-rw-r--r--  1 jenkins jenkins 4.7M Sep  2 13:10 winstone14342431687842909196.jar
-r--r--r--  1 msis    msis      11 Sep  2 13:10 .X0-lock
-r--r--r--  1 gdm     gdm       11 Sep  2 13:10 .X1024-lock
-r--r--r--  1 gdm     gdm       11 Sep  2 13:10 .X1025-lock
drwxrwxrwt  2 root    root    4.0K Sep  2 13:10 .X11-unix
-r--r--r--  1 msis    msis      11 Sep  2 13:10 .X1-lock
drwxrwxrwt  2 root    root    4.0K Sep  2 13:10 .XIM-unix
```

4. Which entries were modified recently.

```
msis@master:~/cyber_lab/reports$ ls -lat /tmp | head
total 4892
drwxrwxrwt 20 root    root       4096 Sep  2 13:44 .
drwx------  3 root    root       4096 Sep  2 13:44 systemd-private-d0c941af35fa4
3ea888309d8c88f5abc-fwupd.service-tGmesg
drwx------  4 root    root       4096 Sep  2 13:11 snap-private-tmp
drwxrwxrwt  2 root    root       4096 Sep  2 13:10 .X11-unix
-r--r--r--  1 msis    msis         11 Sep  2 13:10 .X1-lock
-r--r--r--  1 msis    msis         11 Sep  2 13:10 .X0-lock
drwxrwxrwt  2 root    root       4096 Sep  2 13:10 .ICE-unix
drwx------  3 root    root       4096 Sep  2 13:10 systemd-private-d0c941af35fa4
3ea888309d8c88f5abc-colord.service-99VIOW
-r--r--r--  1 gdm     gdm          11 Sep  2 13:10 .X1025-lock
```

Why can temporary directories be relevant during a cybersecurity investigation?

Temporary directories can contain suspicious scripts, downloaded files, malware, or temporary data created during an attack.

Exercise 14 – Explore the `/proc` Filesystem

Investigate:

/proc

Examine:

cat /proc/cpuinfo

cat /proc/meminfo

Then identify your current shell PID:

echo $$

```
msis@master:~/cyber_lab/reports$ echo $$
8592
```

Check whether a corresponding directory exists under `/proc`.
Information about current shell process

```
msis@master:~/cyber_lab/reports$ ls -ld /proc/$$
dr-xr-xr-x 9 msis msis 0 Sep  2 13:25 /proc/8592
```

1. What information does `/proc/cpuinfo` provide?

It provides information about the system's processors and CPU features.

2. What information does `/proc/meminfo` provide?

It provides information about system memory and memory usage.

3. Why do directories with numbers appear inside `/proc`?

Numbered directories represent running processes, and each number is a Process ID (PID).

4. Is `/proc` an ordinary disk directory?

No. /proc is a virtual filesystem created by the Linux kernel to provide system and process information.

5. Why could `/proc` be useful during security investigation?

/proc can provide information about running processes, system activity, memory, and process details that may help identify suspicious activity.

Exercise 15 – Explore `/dev`

/dev

Examine:

ls -l /dev/null

ls -l /dev/zero

```
msis@master:~/cyber_lab/reports$ ls -l /dev/null
crw-rw-rw- 1 root root 1, 3 Sep  2 13:10 /dev/null
msis@master:~/cyber_lab/reports$ ls -l /dev/zero
crw-rw-rw- 1 root root 1, 5 Sep  2 13:10 /dev/zero
```

1. What does `/dev` represent?

/dev contains special files that provide access to devices and hardware in Linux.

2. Are the entries ordinary user files?

No. Many entries in /dev are special device files rather than ordinary user files

3. Determine the file type of `/dev/null`.

/dev/null is a character special device

4. Why does Linux expose devices through the filesystem?

Linux exposes devices through the filesystem so programs can interact with hardware and system devices using standard file operations.

```
msis@master:~/cyber_lab/reports$ #1
msis@master:~/cyber_lab/reports$ ls -la /dev |head -5
total 4
drwxr-xr-x  20 root root            4980 Sep  2 13:10 .
drwxr-xr-x  24 root root            4096 Aug 11 13:50 ..
crw-------   1 root root       10,   262 Sep  2 13:10 acpi_thermal_rel
crw-r--r--   1 root root       10,   235 Sep  2 13:10 autofs
msis@master:~/cyber_lab/reports$ #2
msis@master:~/cyber_lab/reports$ ls -l /dev/null /dev/zero
crw-rw-rw- 1 root root 1, 3 Sep  2 13:10 /dev/null
crw-rw-rw- 1 root root 1, 5 Sep  2 13:10 /dev/zero
msis@master:~/cyber_lab/reports$ #3
msis@master:~/cyber_lab/reports$ file /dev/null
/dev/null: character special (1/3)
msis@master:~/cyber_lab/reports$ #4
msis@master:~/cyber_lab/reports$ ls -l /dev |head -5
total 0
crw-------   1 root root       10,   262 Sep  2 13:10 acpi_thermal_rel
crw-r--r--   1 root root       10,   235 Sep  2 13:10 autofs
drwxr-xr-x   2 root root            720 Sep  2 13:10 block
crw-------   1 root root       10,   234 Sep  2 13:10 btrfs-control
```

Exercise 16 – Filesystem Space Investigation

Determine:

1. Total filesystem capacity.

2. Used space.

3. Available space.

4. Percentage of filesystem utilization.

```
msis@master:~/cyber_lab/reports$ #1#2#3#4
msis@master:~/cyber_lab/reports$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           1.6G  2.8M  1.6G   1% /run
/dev/nvme0n1p7  233G   64G  158G  29% /
tmpfs           7.7G  4.0K  7.7G   1% /dev/shm
tmpfs           5.0M  8.0K  5.0M   1% /run/lock
efivarfs        246K  180K   62K  75% /sys/firmware/efi/efivars
tmpfs           7.7G     0  7.7G   0% /run/qemu
/dev/nvme0n1p1  196M   77M  120M  40% /boot/efi
tmpfs           1.6G  2.5M  1.6G   1% /run/user/1000
```

5. Space occupied by your home directory.

6. Space occupied by the `cyber_lab` directory.

```
msis@master:~$ #5
msis@master:~$ du -sh ~
45G     /home/msis
msis@master:~$ #6
msis@master:~$ du -sh cyber_lab
48K     cyber_lab
```

Use appropriate combinations of:
df
df -h
du
du -h
du -sh

Explain the difference between `df` and `du`.

df tells youhowmuchspacethefilesystemhasand how much is free; **du** tells you where that space is being used.

Exercise 17 – Cybersecurity Investigation Scenario

Assume that you are investigating a Linux machine after suspicious activity has been reported.

You are initially asked to examine only the filesystem.

Identify which of the following locations you would investigate and explain why:

Location	Incestigation	Why	Security relevance
/home	Yes	Direct evidence of user activity	.bash_history, SSH keys, downloaded tools, browser artifacts
/etc	Yes	Defines what the system is set up to allow	Unauthorized accounts, sudoers rules, cron jobs, SSH config changes
/var/log	Yes	Primary timeline of what already happened	Login history, failed logins, auth logs, install logs
/tmp	Yes	World-writable, wiped on reboot — malware staging spot	Unexpected scripts/binaries, hidden files, mismatched ownership
/proc	Yes	Live view of what's happening right now	Real cmdline, actual running binary (exe), open network sockets
/dev	Sometimes	Normally routine, but can be abused	Unexpected device nodes, data disguised as a device file
/root	If escalation suspected	Shows what an attacker did with elevated access	Root's shell history, SSH keys, config changes as superuser